

Alex Levesque

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EDUCATION

Queen’s University <i>BASc Applied Mathematics and Computer Engineering GPA: 3.80/4.00</i>	Kingston, ON <i>Sep 2024 – Apr 2028</i>
• Relevant Coursework: Data Structures, OOP, Real Analysis, Linear Algebra, Probability, Microeconomics	
University of Warwick <i>Incoming Exchange Semester</i>	Coventry, UK <i>Jan 2027 – Apr 2027</i>
• Potential Coursework: Data Science, Financial Markets, Time Series Analysis, Econometrics	

EXPERIENCE

Data Science Intern <i>Scotiabank</i>	May 2026 – Aug 2026 <i>Toronto, ON</i>
• Incoming Summer 2026 , expected responsibilities include building data pipelines, developing predictive models using Python and Spark, and creating PowerBI dashboards to translate complex data into actionable insights	
LLM Researcher <i>Algoverse AI Research</i>	Jun 2025 – Nov 2025 <i>Remote</i>
• Conducted AI research under mentorship from PhDs at Meta, OpenAI, and Princeton, exploring novel contrastive decoding methods that improved efficiency by 2% across MoE benchmarks	
• Implemented these methods in large language models (Qwen3-235B-A22B, GPT-OSS) by designing a self-contrastive Mixture-of-Experts architecture using PyTorch and Jupyter Notebooks	
Generative AI Engineering Intern <i>Government of Canada</i>	May 2025 – Aug 2025 <i>Ottawa, ON</i>
• Designed and implemented Azure-based GenAI workflows to transcribe, categorize, and organize refugee documents achieving 93% accuracy , reducing manual processing time, and improving operational efficiency	
• Engineered an Azure chatbot leveraging an ETL pipeline, Cosmos DB (NoSQL), sentiment analysis, and vector storage to automate JSON legal document classification and presented results to stakeholders	

AWARDS

GRAMMY Music Award , The Recording Academy	Feb 2024
• Received a GRAMMY Award for contributions to Killer Mike’s <i>MICHAEL</i> , 2024 Rap Album of the Year	
• Produced for the Migos, Gunna, Peso Pluma, Roddy Ricch, Polo G, A Boogie, Lil Tjay, Skepta, Cam’ron	
Canada’s Top Students , Scotiabank	Nov 2025
• Selected as 1 of 30 attendees for Scotiabank’s Canada’s Top Students conference and case competition	

PROJECTS

For more projects and in-depth analyses regarding my projects, click [here](#).

Adaptive Reinforcement Learning Ensemble Strategy , Queen’s AI Club
• Implemented and trained an ensemble of PPO, A2C, and TD3 reinforcement learning agents for systematic portfolio allocation and position sizing across market regimes, with experiment tracking and version control via Git
• Designed a risk-aware diversification and rebalancing framework that reduced portfolio volatility and improved risk-adjusted returns, validated via Sharpe ratio, max drawdown, and turnover analysis against a SPY buy-and-hold
Autonomous Multi-Agent Robotic Firefighting
• Developed and optimized Lloyd’s algorithm in MATLAB for adaptive k-means clustering of dynamic fire hotspot data, reducing cluster convergence time by 30% and improving autonomous robot deployment efficiency
• Leveraged GIS wildfire spatial data and analysis tools to delineate and model fire perimeters, enabling accurate real-time input for robotic firefighting cluster optimization and enhancing fire containment strategies

SKILLS

Programming: Python, C++, C, Java, Assembly x86, SQL, Pandas, PyTorch, Scikit-learn, Jupyter Notebook
Tools & Platforms: Microsoft Azure, Git, Copilot Studio, MATLAB, n8n, Firebase, Solidworks, Excel
Interests: Native Bilingual (English and French), Karate Black Belt, Pro-Bono Web Dev, Team Sports